

COMBINATORICS

ASSIGNMENT 5

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1. PROBLEM 1

In how many ways can we build a 7-beaded circular necklace if there are 5 available types of gemstones (repetition) allowed and all rotations and reflections of a given necklace are equivalent. What if we have q gemstones?

Solution. □

2. PROBLEM 2

A rectangular prism has equilateral triangles for its base and top, and three rectangular lateral faces. In how many ways can we color if the faces can be colored using any one of the 5 colors.

Solution. □

3. PROBLEM 3

Find the pattern inventory of the red-blue vertex colorings of a tetrahedron. Repeat the problem with red-blue edge colorings.

Solution. □

4. PROBLEM 4

How many 10 red, 12 blue, 3 green squares are there in the coloring of a 5×5 chessboard?

Solution. □

5. PROBLEM 5

Find the pattern inventory of the red-blue vertex colorings of a floating cube (i.e. the cube can freely move in space).

Solution. □